

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Clarification for Gen 4 Exit Flow from CL1 or CL2 State
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

This ECN replaces the misleading references to the SB TxFFE registers with a reference to the Gen 4 behavior in Training state
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Benefits as a result of the changes:

This ECN adds clarity to the spec

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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NA

An analysis of the hardware implications:
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NA

An analysis of the software implications:
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NA

An analysis of the compliance testing implications:
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NA

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Actual Change

(a). Section 4.2.1.6.5.4 Gen 4 Exit Flow from CL1 or CL2 State

To Text:

A USB4 Port may initiate exit from CL1 or CL2 state due to an internal objection or detection of LFPS on its Lane 0 receiver. When initiating exit from CL1 or CL2 state, a USB4 Port shall:

1. Send a Low Frequency Periodic Signaling (LFPS) burst on Lane 0 until all the following conditions are met:
 - If the exit was initiated by detecting LFPS:
 - The receiver no longer detects LFPS.
 - The transmitter sent at least 16 LFPS cycles.
 - Else:
 - The receiver detected LFPS.
 - After the receiver detected LFPS, the transmitter sent at least 16 LFPS cycles.

The Port shall stop sending LFPS within tStopLFPS2 time after the conditions are met.

If the exit from CL1 or CL2 was initiated by detecting LFPS on the Lane 0 receiver, the Port shall transmit the LFPS on the Lane 0 transmitter within tWarmupCL1 or tWarmupCL2 respectively.

If the receiver did not detect LFPS after tTrainingAbort2 time, the Router shall initiate a Disconnect by driving SBTX to a logical low state for tDisconnectTx.

2. Return to Electrical Idle for tPreData (see Table 3-33).
3. Transition to the Training.TS1 sub-state using the TxFFE Preset that was used prior to CL1/2 entry.
4. Each Lane Adapter shall complete Symbol lock within tRxLock time and proceed ~~according to Table 4-22 and Table 4-23 as described in 4.2.1.3.2.2,~~ without performing the TxFFE negotiation flow, until the transition to CL0 state.
5. The first RS-FEC block shall be a De-skew Block as defined in Section 4.4.4.